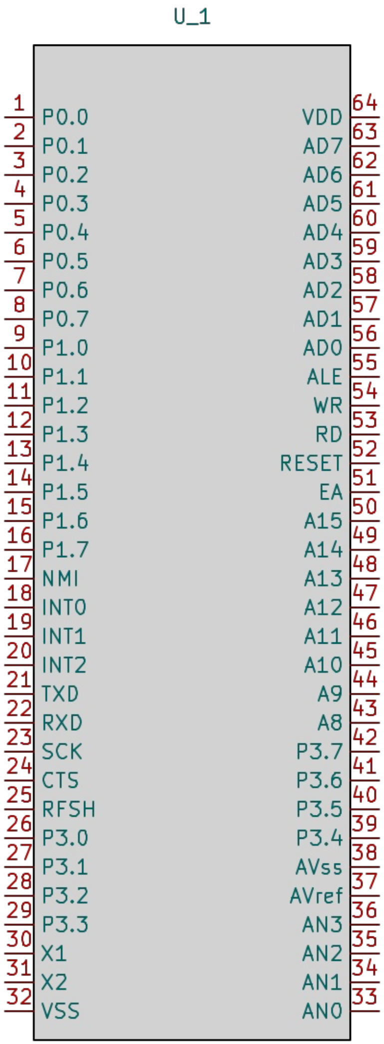
xD-69 Hardware Description

This amazing logo was designed by someone called kilésengati on Discord :)

CPU: NEC uPD78312 – 8/16 bit embedded microcontroller



NEC 78k/3 series microcontroller for embedded devices. CISC(Complex instruction set computer) processor, little endian.
No current ROM dump. Has a PROM that contains opcode stuff I think
Serial MIDI transmission uses RXD and TXD CPU pins.

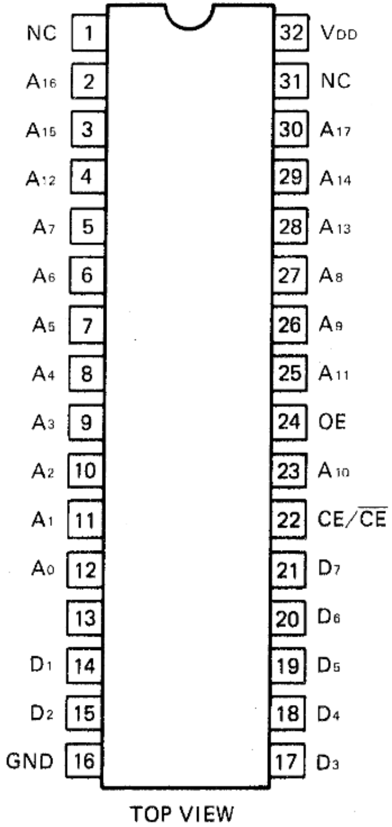
Pin No.	Pin Name	Input/Output	Description	Pin No.	Pin Name	Input/Output	Description
1	P0.0	O	Port 0	33	AN0	I	Analog input
2	P0.1	O	Port 0	34	AN1	I	Analog input
3	P0.2	O	Port 0	35	AN2	I	Analog input
4	P0.3	O	Port 0	36	AN3	I	Analog input
5	P0.4	O	Port 0	37	AVREF	O	Reference voltage
6	P0.5	O	Port 0	38	AVSS	-	Analog VSS
7	P0.6	O	Port 0	39	P3.4	I/O(NC)	Port 3
8	P0.7	O	Port 0	40	P3.5	I/O(NC)	Port 3
9	P1.0	I/O	Port 1	41	P3.6	I/O	Port 3
10	P1.1	I/O	Port 1	42	P3.7	I/O	Port 3
11	P1.2	I/O	Port 1	43	A8	O	Address bus
12	P1.3	I/O	Port 1	44	A9	O	Address bus
13	P1.4	I/O	Port 1	45	A10	O	Address bus
14	P1.5	I/O	Port 1	46	A11	O	Address bus
15	P1.6	I/O	Port 1	47	A12	O	Address bus
16	P1.7	I/O(NC)	Port 1	48	A13	O	Address bus
17	NMI	I	Non-maskable interrupt	49	A14	O	Address bus
18	INT0	I	Interrupt from externals	50	A15	O	Address bus
19	INT1	I	Interrupt from externals	51	$\overline{\text{EA}}$	I	External access
20	INT2	I	Interrupt from externals	52	$\overline{\text{RESET}}$	I	Reset
21	TXD	O	Transfer serial data	53	$\overline{\text{RD}}$	O	Read strobe
22	RXD	I	Receive serial data	54	$\overline{\text{WR}}$	O	Write strobe
23	$\overline{\text{SCK}}$	O(NC)	Serial transmit clock output	55	ALE	O	Address latch enable
24	$\overline{\text{CTS}}$	I/O	Clear to send	56	AD0	I/O	Address/data bus
25	$\overline{\text{RFSH}}$	O(NC)	Refresh	57	AD1	I/O	Address/data bus
26	P3.0	I	Port 3	58	AD2	I/O	Address/data bus
27	P3.1	I	Port 3	59	AD3	I/O	Address/data bus
28	P3.2	I	Port 3	60	AD4	I/O	Address/data bus
29	P3.3	I	Port 3	61	AD5	I/O	Address/data bus
30	X1	I	Crystal	62	AD6	I/O	Address/data bus
31	X2	I	Crystal	63	AD7	I/O	Address/data bus
32	VSS	-	Ground	64	VDD	-	

Read-Only Memories

PCM storage ROM – contains PCM partials for the preset playback. Has been ROM dumped. Roland owns the samples so you need to provide your own copy

OS ROM - This is where the D-50 runs its code from for the D-50 operating system. I have provided the disassembly in another folder.
I managed to disassemble the binary using MAME disasm command line

PCM ROM A/B
TC532000



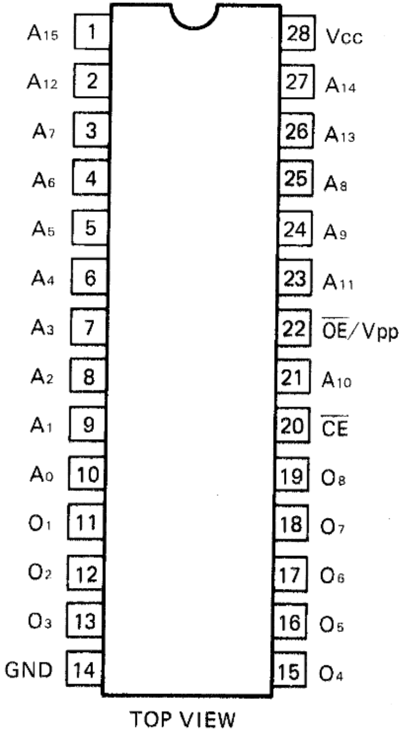
TC532000 ROM(PCM)

Pin No.	Pin Name	Input/Output	Description
1	NC	-	No connection
2	A16	I	Address inputs
3	A15	I	Address inputs
4	A12	I	Address inputs
5	A7	I	Address inputs
6	A6	I	Address inputs
7	A5	I	Address inputs
8	A4	I	Address inputs
9	A3	I	Address inputs
10	A2	I	Address inputs
11	A1	I	Address inputs
12	A0	I	Address inputs
13	D0	O	Data outputs
14	D1	O	Data outputs
15	D2	O	Data outputs
16	GND	-	Ground
17	D3	O	Data outputs
18	D4	O	Data outputs
19	D5	O	Data outputs
20	D6	O	Data outputs
21	D7	O	Data outputs
22	CE/$\overline{CE}$	I	Chip enable input
23	A10	I	Address inputs
24	$\overline{OE}$	I	Output enable input
25	A11	I	Address inputs
26	A9	I	Address inputs
27	A8	I	Address inputs
28	A13	I	Address inputs
29	A14	I	Address inputs
30	A17	I	Address inputs
31	NC	-	No connection
32	VDD	-	Power

MBM27C512 ROM (OS)

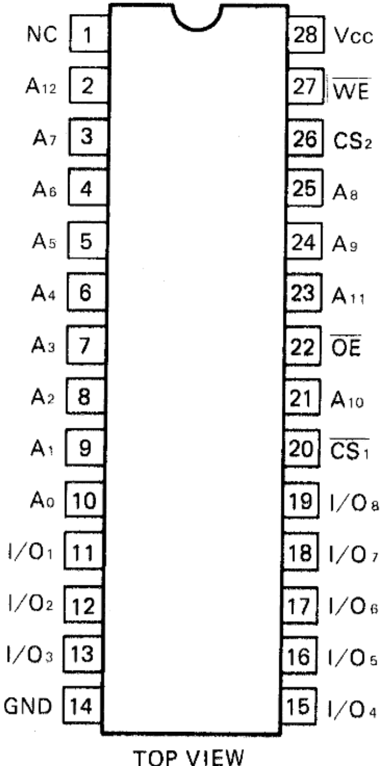
Pin No.	Pin Name	Input/Output	Description
1	A15	I	Address input
2	A12	I	Address input
3	A7	I	Address input
4	A6	I	Address input
5	A5	I	Address input
6	A4	I	Address input
7	A3	I	Address input
8	A2	I	Address input
9	A1	I	Address input
10	A0	I	Address input
11	O1	O	Data output
12	O2	O	Data output
13	O3	O	Data output
14	GND	-	Ground
15	O4	O	Data output
16	O5	O	Data output
17	O6	O	Data output
18	O7	O	Data output
19	O8	O	Data output
20	$\overline{CE}$	I	Chip enable input
21	A10	I	Address input
22	$\overline{OE}$/VPP	I	Output enable input
23	A11	I	Address input
24	A9	I	Address input
25	A8	I	Address input
26	A13	I	Address input
27	A14	I	Address input
28	VCC	-	Power

EP ROM MBM27C512

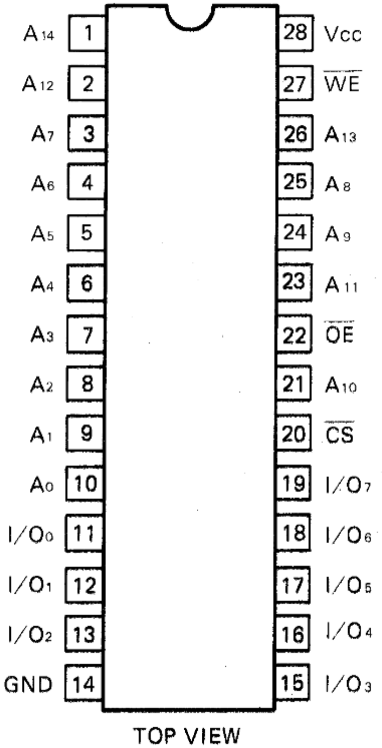


Random-Access Memories

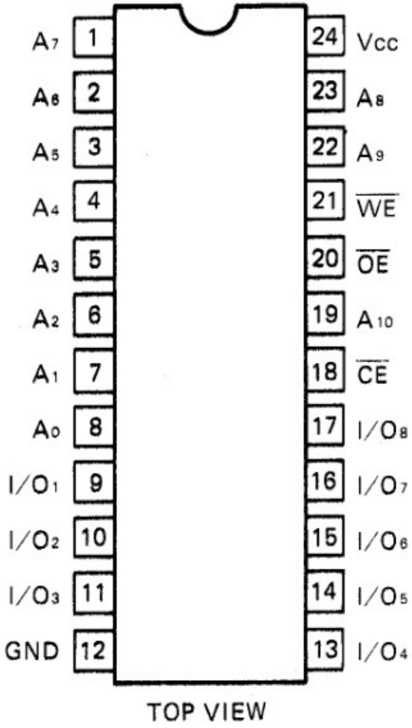
S RAM
HM6264ASP



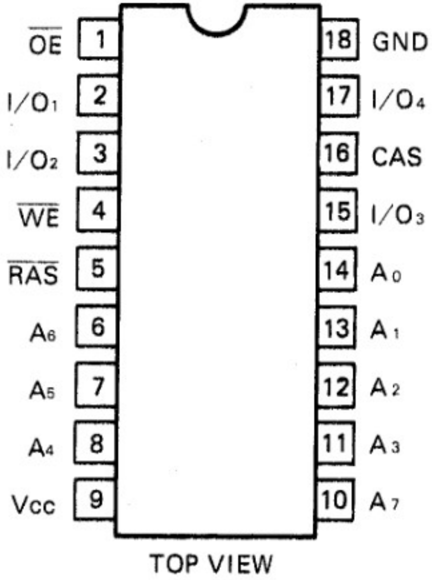
S RAM
HM62256LP



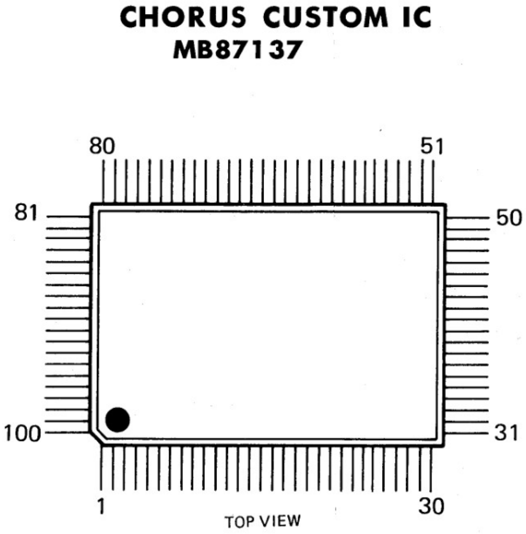
S RAM
LC3517AS



D RAM
μPD41416



Target Custom ASIC – Chorus DSP

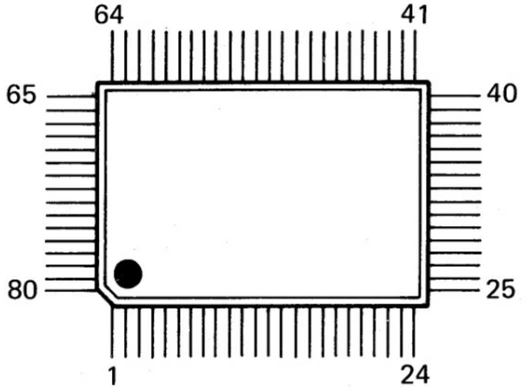


Pin No.	Pin Name	Input/Output	Description	Pin No.	Pin Name	Input/Output	Description
1	RES	I	Reset input	51	O8	O	Data input
2	E	I	Chip enable input	52	O9	O	Data input
3	VDD	-	+5v	53	VDD	-	+5v
4	CS	I	Chip Select input	54	O10	O	Data input
5	$\overline{RW}$	I	Write Pulse input	55	O11	O	Data input
6	$\overline{RD}$	I	Read Pulse input	56	O12	O	Data input
7	$\overline{CS}$	I	Chip select input	57	O13	O	Data input
8	A0	I	Connect to CPU address bus	58	O14	O	Data input
9	A1	I	Connect to CPU address bus	59	O15	O	Data input
10	A2	I	Connect to CPU address bus	60	RA0	O	Connects to SRAM address bus RA13 not used
11	D0	I/O	Connect to CPU data bus	61	$\overline{WE}$	O	SRAM write pulse output
12	D1	I/O	Connect to CPU data bus	62	RA1	O	Connects to SRAM address bus RA13 not used
13	D2	I/O	Connect to CPU data bus	63	RA2	O	Connects to SRAM address bus RA13 not used
14	D3	I/O	Connect to CPU data bus	64	RA3	O	Connects to SRAM address bus RA13 not used
15	VSS	-	GND	65	VSS	-	GND
16	D4	I/O	Connect to CPU data bus	66	RA4	O	Connects to SRAM address bus RA13 not used
17	D5	I/O	Connect to CPU data bus	67	RA5	O	Connects to SRAM address bus RA13 not used
18	D6	I/O	Connect to CPU data bus	68	RA6	O	Connects to SRAM address bus RA13 not used
19	D7	I/O	Connect to CPU data bus	69	RA7	O	Connects to SRAM address bus RA13 not used
20	$\overline{DOE}$	I	Data out enable input	70	RA8	O	Connects to SRAM address bus RA13 not used
21	INCK	I	Input data latch clock input	71	$\overline{OE}$	O	SRAM out enable output
22	SIN	I	Sync input	72	RA9	O	Connects to SRAM address bus RA13 not used
23	SOUT	O	Sync output	73	RA10	O	Connects to SRAM address bus RA13 not used
24	LRS	I	L/R select input	74	RA11	O	Connects to SRAM address bus RA13 not used
25	I0	I	Data input	75	$\overline{CE}$	O	SRAM chip enable output
26	I1	I	Data input	76	RA12	O	Connects to SRAM address bus RA13 not used
27	I2	I	Data input	77	PD7	I/O	Connect to SRAM data bus
28	VDD	-	+5v	78	VDD	-	+5v
29	I3	I	Data input	79	RA13	O	Connects to SRAM address bus RA13 not used
30	I4	I	Data input	80	PD6	I/O	Connect to SRAM data bus
31	I5	I	Data input	81	PD5	I/O	Connect to SRAM data bus
32	I6	I	Data input	82	PD4	I/O	Connect to SRAM data bus
33	I7	I	Data input	83	PD3	I/O	Connect to SRAM data bus
34	I8	I	Data input	84	PD2	I/O	Connect to SRAM data bus
35	I9	I	Data input	85	PD1	I/O	Connect to SRAM data bus
36	I10	I	Data input	86	PD0	I/O	Connect to SRAM data bus
37	I11	I	Data input	87	VSS	-	GND
38	I12	I	Data input	88	X1	I	Master clock input
39	I13	I	Data input	89	X2	O	not used
40	VSS	-	GND	90	VSS	-	GND
41	I14	I	Data input	91	ROMT	I	No description
42	I15	I	Data input	92	RAMT	I	No description
43	O0	O	Data input	93	CTRT	I	No description
44	O1	O	Data input	94	THRU	I	No description
45	O2	O	Data input	95	ECTL	I	External control select input
46	O3	O	Data input	96	ADDA	I	No description
47	O4	O	Data input	97	OFST	I	offset binary select input
48	O5	O	Data input	98	PSFT	I	No description
49	O6	O	Data input	99	LHLD	O	Signal output for S/H – not used
50	O7	O	Data input	100	RHLD	O	Signal output for S/H – not used

Please see:
https://github.com/pxlsr/FujiCells
for a detailed and comprehensive documentation of the ASICs

Target Custom ASIC – Reverb DSP

REVERB CUSTOM IC
MB87126-006



TOP VIEW

Pin No.	Pin Name	Input/Output	Description
1	DC0	O	Data output for chorus chip and DAC
2	DC1	O	Data output for chorus chip and DAC
3	STRT	I	Pulled low
4	DIN	I	Pulled low
5	CLEA	I	Pulled low
6	RD0	O	Control output for enable and for S/H and Lower four bit D/A conversion
7	RD1	O	Control output for enable and for S/H and Lower four bit D/A conversion
8	RD2	O	Control output for enable and for S/H and Lower four bit D/A conversion
9	RD3	O	Control output for enable and for S/H and Lower four bit D/A conversion
10	RD4	O	Control output for enable and for S/H and Lower four bit D/A conversion
11	RSET	I	Pulled low
12	VSS	-	GND
13	SLRQ	I	Pulled low
14	MSCK	I	Master clock input
15	VSS	-	GND
16	SLCK	O	Not used
17	TEST	I	Pulled low
18	TMB	O	Time base signal output
19	VDD	-	+5V
20	$\overline{\text{LOAD}}$	O	Sync signal output
21	SYNC	I	Sync signal input
22	INCK	I	Data latch clock input for initialization
23	ERCL	I	Busy reset output
24	BUSY	O	Serial data transfer error output (parity check)
25	SXD	I	Serial data input
26	SCK	I	Serial data read-in clock input
27	DA0	O	Connect to RAM address bus
28	DA1	O	Connect to RAM address bus
29	DA2	O	Connect to RAM address bus
30	DA3	O	Connect to RAM address bus
31	DA4	O	Connect to RAM address bus
32	DA5	O	Connect to RAM address bus
33	VDD	-	+5V
34	DA6	O	Connect to RAM address bus
35	DA7	O	Connect to RAM address bus
36	VSS	-	GND
37	$\overline{\text{RAS}}$	O	Row address strobe output
38	$\overline{\text{WE}}$	O	DRAM write pulse output
39	$\overline{\text{CAS}}$	O	Column address strobe output
40	DR0	I/O	Connect to RAM data bus, Synth and Chorus data input
41	DR1	I/O	Connect to RAM data bus, Synth and Chorus data input
42	DR2	I/O	Connect to RAM data bus, Synth and Chorus data input
43	DR3	I/O	Connect to RAM data bus, Synth and Chorus data input
44	DR4	I/O	Connect to RAM data bus, Synth and Chorus data input
45	DR5	I/O	Connect to RAM data bus, Synth and Chorus data input
46	DR6	I/O	Connect to RAM data bus, Synth and Chorus data input
47	DR7	I/O	Connect to RAM data bus, Synth and Chorus data input
48	DR8	I/O	Connect to RAM data bus, Synth and Chorus data input
49	DR9	I/O	Connect to RAM data bus, Synth and Chorus data input
50	DR10	I/O	Connect to RAM data bus, Synth and Chorus data input
51	DR11	I/O	Connect to RAM data bus, Synth and Chorus data input
52	VSS	-	GND
53	DR12	I/O	Connect to RAM data bus, Synth and Chorus data input
54	DR13	I/O	Connect to RAM data bus, Synth and Chorus data input
55	DR14	I/O	Connect to RAM data bus, Synth and Chorus data input
56	DR15	I/O	Connect to RAM data bus, Synth and Chorus data input
57	DR16	I/O	Connect to RAM data bus, Synth and Chorus data input
58	DR17	I/O	Connect to RAM data bus, Synth and Chorus data input
59	DR18	I/O	Connect to RAM data bus, Synth and Chorus data input
60	DR19	I/O	Connect to RAM data bus, Synth and Chorus data input
61	DR20	I/O	Connect to RAM data bus, Synth and Chorus data input
62	DR21	I/O	Connect to RAM data bus, Synth and Chorus data input
63	DR22	I/O	Connect to RAM data bus, Synth and Chorus data input
64	DR23	I/O	Connect to RAM data bus, Synth and Chorus data input
65	VSS	-	GND
66	DC2	O	Data output for chorus chip and DAC
67	DC3	O	Data output for chorus chip and DAC
68	DC4	O	Data output for chorus chip and DAC
69	DC5	O	Data output for chorus chip and DAC
70	DC6	O	Data output for chorus chip and DAC
71	DC7	O	Data output for chorus chip and DAC
72	DC8	O	Data output for chorus chip and DAC
73	VDD	-	+5V
74	DC9	O	Data output for chorus chip and DAC
75	DC10	O	Data output for chorus chip and DAC
76	DC11	O	Data output for chorus chip and DAC
77	DC12	O	Data output for chorus chip and DAC
78	DC13	O	Data output for chorus chip and DAC
79	DC14	O	Data output for chorus chip and DAC
80	DC15	O	Data output for chorus chip and DAC

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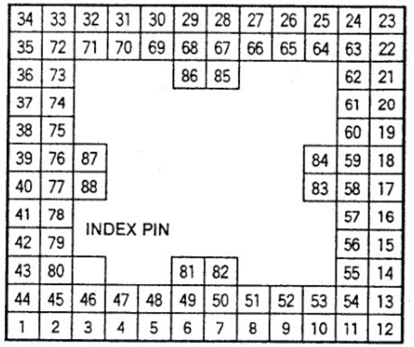
Target Custom ASIC – Synth (DSP core)

SYNTHE CUSTOM IC
MB87136



TOP VIEW

Pin No.	Pin Name	Input/Output	Description
1	$\overline{CS}$	I	Chip select
2	A0	I	Connect to CPU address bus
3	A1	I	Connect to CPU address bus
4	A2	I	Connect to CPU address bus
5	A3	I	Connect to CPU address bus
6	A4	I	Connect to CPU address bus
7	D0	I/O	Connect to CPU data bus
8	D1	I/O	Connect to CPU data bus
9	D2	I/O	Connect to CPU data bus
10	D3	I/O	Connect to CPU data bus
11	PD0	I	Connect to ROM data bus
12	PD1	I	Connect to ROM data bus
13	PD2	I	Connect to ROM data bus
14	PD3	I	Connect to ROM data bus
15	RA0	O	Connect to ROM address bus
16	RA1	O	Connect to ROM address bus
17	RA2	O	Connect to ROM address bus
18	RA3	O	Connect to ROM address bus
19	RA4	O	Connect to ROM address bus
20	RA5	O	Connect to ROM address bus
21	RA6	O	Connect to ROM address bus
22	RA7	O	Connect to ROM address bus
23	RA8	O	Connect to ROM address bus
24	RA9	O	Connect to ROM address bus
25	RA10	O	Connect to ROM address bus
26	RA11	O	Connect to ROM address bus
27	O0	O	Data output
28	O1	O	Data output
29	O2	O	Data output
30	O3	O	Data output
31	O4	O	Data output
32	O5	O	Data output
33	O6	O	Data output
34	O7	O	Data output
35	O8	O	Data output
36	SH0	O	Not used
37	SH1	O	Not used
38	-	-	Not used
39	X1	I/O	Crystal input (32.768mhz)
40	32M	O	The same frequency as master clock
41	-	-	-
42	$\overline{SYI}$	I	Sync signal input
43	$\overline{WR}$	I	Write pulse input
44	$\overline{INT}$	O	Interrupt output
45	$\overline{OE}$	I	Output enable input

Pin No.	Pin Name	Input/Output	Description
46	A5	I	Connect to CPU address bus
47	A6	I	Connect to CPU address bus
48	A7	I	Connect to CPU address bus
49	A8	I	Connect to CPU address bus
50	D4	I/O	Connect to CPU data bus
51	D5	I/O	Connect to CPU data bus
52	D6	I/O	Connect to CPU data bus
53	D7	I/O	Connect to CPU data bus
54	PD4	I	Connect to ROM data bus
55	PD5	I	Connect to ROM data bus
56	PD6	I	Connect to ROM data bus
57	PD7	I	Connect to ROM data bus
58	RA12	O	Connect to ROM address bus
59	RA13	O	Connect to ROM address bus
60	RA14	O	Connect to ROM address bus
61	RA15	O	Connect to ROM address bus
62	RA16	O	Connect to ROM address bus
63	RA17	O	Connect to ROM address bus
64	RA18	O	Connect to ROM address bus
65	RA19	O	Connect to ROM address bus
66	O9	O	Data output
67	O10	O	Data output
68	O11	O	Data output
69	O12	O	Data output
70	O13	O	Data output
71	O14	O	Data output
72	O15	O	Data output
73	SH2	O	Not used
74	SH3	O	Not used
75	-	-	Not used
76	X2	I/O	Crystal input
77	16M	O	Output frequency is half of master clock
78	CKIN	I	Input frequency is a combination of the master clock and one half of master clock
79	-	-	Not used
80	$\overline{RD}$	I	Read pulse input
81	VSS	-	GND
82	VDD	-	+5V
83	VDD	-	+5V
84	VSS	-	GND
85	VSS	-	GND
86	VDD	-	+5V
87	VDD	-	+5V
88	VSS	-	GND

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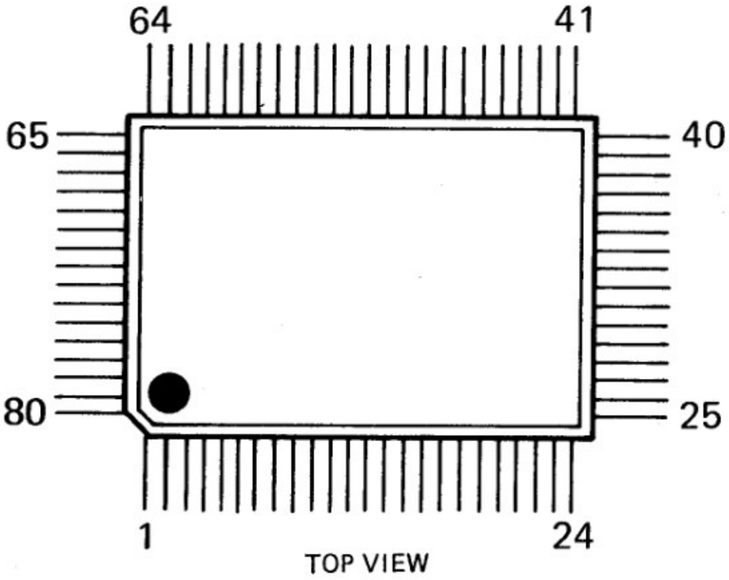
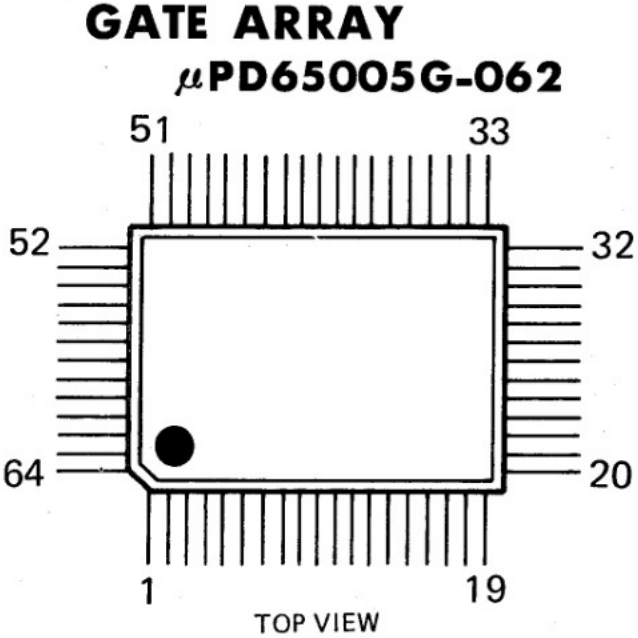
Target Custom Gate Array – LCD controller with possible peripheral control logic?

GATE ARRAY
HG61H25B18F



PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O
1	SYNT2	O (NC)	21	ALE	I	41	EC	O	61	R2	I
2	IRAM	O (NC)	22	$\overline{WR}$	I	42	O0	O	62	R3	I
3	RAM	O	23	$\overline{RD}$	I	43	O1	O	63	R4	I
4	A7	O	24	$\overline{RESET}$	I	44	O2	O	64	R5	I
5	A6	O	25	A15	I	45	O3	O	65	R6	I
6	A5	O	26	A14	I	46	O4	O	66	R7	I
7	A4	O	27	A13	I	47	O5	O	67	CORUS	O
8	A3	O	28	A12	I	48	O6	O	68	SCK	O
9	A2	O	29	A11	I	49	O7	O	69	SXD	O
10	A1	O	30	A10	I	50	S0	O	70	BUSY	I
11	A0	O	31	A9	I	51	S1	O	71	ERCL	O
12	Vss	-	32	A8	I	52	Vss	-	72	LOAD	I
13	AD7	I/O	33	VDD	-	53	S2	O	73	VDD	-
14	AD6	I/O	34	ARS	I (HIGH)	54	S3	O	74	TMB	I
15	AD5	I/O	35	INT1	O (NC)	55	S4	O	75	SINT1	I (LOW)
16	AD4	I/O	36	INT2	O	56	S5	O	76	SINT2	I (LOW)
17	AD3	I/O	37	DSCAN	O	57	S6	O	77	TEST1	I (LOW)
18	AD2	I/O	38	$\overline{ERAM}$	O	58	S7	O	78	CLK	I
19	AD1	I/O	39	ERAM	O (NC)	59	R0	I	79	TEST2	I (LOW)
20	AD0	I/O	40	RS	O	60	R1	I	80	SYNT1	O

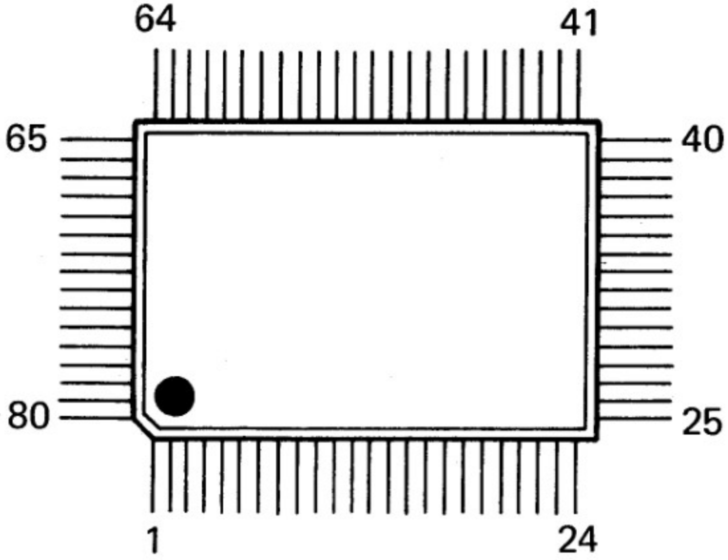
Target Custom Gate Array – Memory card controller



PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O
1	NC	–	17	NC	–	33	NC	–	49	NC	–
2	NC	–	18	NC	–	34	NC	–	50	CD0	I/O
3	AD7	I/O	19	A13	I	35	CA5	O	51	CD1	I/O
4	AD6	I/O	20	A12	I	36	CA6	O	52	CD2	I/O
5	AD5	I/O	21	A11	I	37	CA7	O	53	CD3	I/O
6	AD4	I/O	22	A10	I	38	CA8	O	54	CD4	I/O
7	AD3	I/O	23	A9	I	39	CA9	O	55	CD5	I/O
8	AD2	I/O	24	A8	I	40	CA10	O	56	CD6	I/O
9	AD1	I/O	25	SEL	I (LOW)	41	CA11	O	57	CD7	I/O
10	AD0	I/O	26	Vss	–	42	CA12	O	58	Vss	–
11	Vss	–	27	VDD	–	43	CA13	O	59	VDD	–
12	VDD	–	28	CA0	O	44	CA14	O	60	BATT	I (LOW)
13	ALE	I	29	CA1	O	45	MR	O	61	SENS	I (NC)
14	$\overline{WR}$	I	30	CA2	O	46	CWR	O	62	RCS	I
15	$\overline{RD}$	I	31	CA3	O	47	CCS	O	63	$\overline{CS}$	I
16	A14	I	32	CA4	O	48	CRD	O	64	NC	–

Target Custom Gate Array – Keyboard controller

GATE ARRAY
MB63H149



TOP VIEW

PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O	PIN NO.	NAME	I/O
1	T7	O	21	BR9	I	41	AD7	I/O	61	RA1	O
2	BR0	I	22	MK9	I	42	CA8	I	62	RA10	O
3	MK0	I	23	BR10	I	43	CA9	I	63	RA2	O
4	BR1	I	24	MK10	I	44	CA10	I (LOW)	64	$\overline{\text{ROE}}$	I/O
5	MK1	I	25	RES	I	45	CS	I	65	RA3	O
6	BR2	I	26	$\overline{\text{EXCK}}$	I/O	46	XT1	I	66	$\overline{\text{RWE}}$	O
7	MK2	I	27	E	I (HIGH)	47	XT2	O (NC)	67	RA4	O
8	BR3	I	28	INT	O	48	ASEL	O (NC)	68	RA9	O
9	MK3	I	29	AS	I	49	MOD1	I (HIGH)	69	RA5	O
10	BR4	I	30	$\overline{\text{CRES}}$	O (NC)	50	MOD2	I (LOW)	70	RA8	O
11	MK4	I	31	$\overline{\text{CRNW}}$	I	51	RD3	I/O	71	RA6	O
12	Vss	-	32	SRCK	O (NC)	52	Vss	-	72	RA7	O
13	BR5	I	33	VDD	-	53	RD4	I/O	73	VDD	-
14	MK5	I	34	AD0	I/O	54	RD2	I/O	74	T0	O
15	BR6	I	35	AD1	I/O	55	RD5	I/O	75	T1	O
16	MK6	I	36	AD2	I/O	56	RD1	I/O	76	T2	O
17	BR7	I	37	AD3	I/O	57	RD6	I/O	77	T3	O
18	MK7	I	38	AD4	I/O	58	RD0	I/O	78	T4	O
19	BR8	I	39	AD5	I/O	59	RD7	I/O	79	T5	O
20	MK8	I	40	AD6	I/O	60	RA0	O	80	T6	O